

1. Eliminate the parameter and draw the parametrized curve.

(a) $(x, y) = (6 - 4t, -1 + 3t)$

$$x = 6 - 4t \rightarrow t = \frac{x-6}{-4}$$

$$y = -1 + 3t \rightarrow t = \frac{y+1}{3}$$

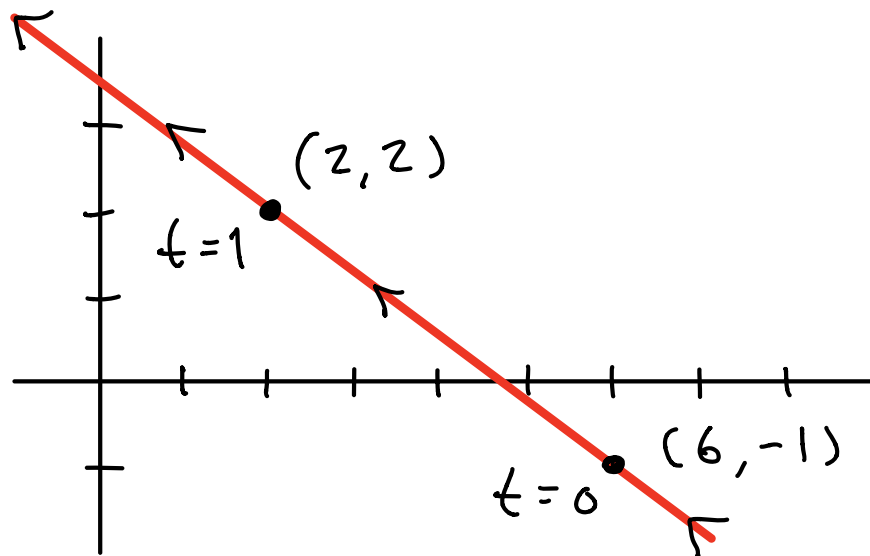
So
$$\frac{y+1}{3} = \frac{x-6}{-4}$$

$$-4y - 4 = 3x - 18$$

$$-4y = 3x - 14$$

$$y = -\frac{3}{4}x + \frac{7}{2}$$

Picture :



$$(b) \quad (x, y) = (e^t, e^{2t} + 1)$$

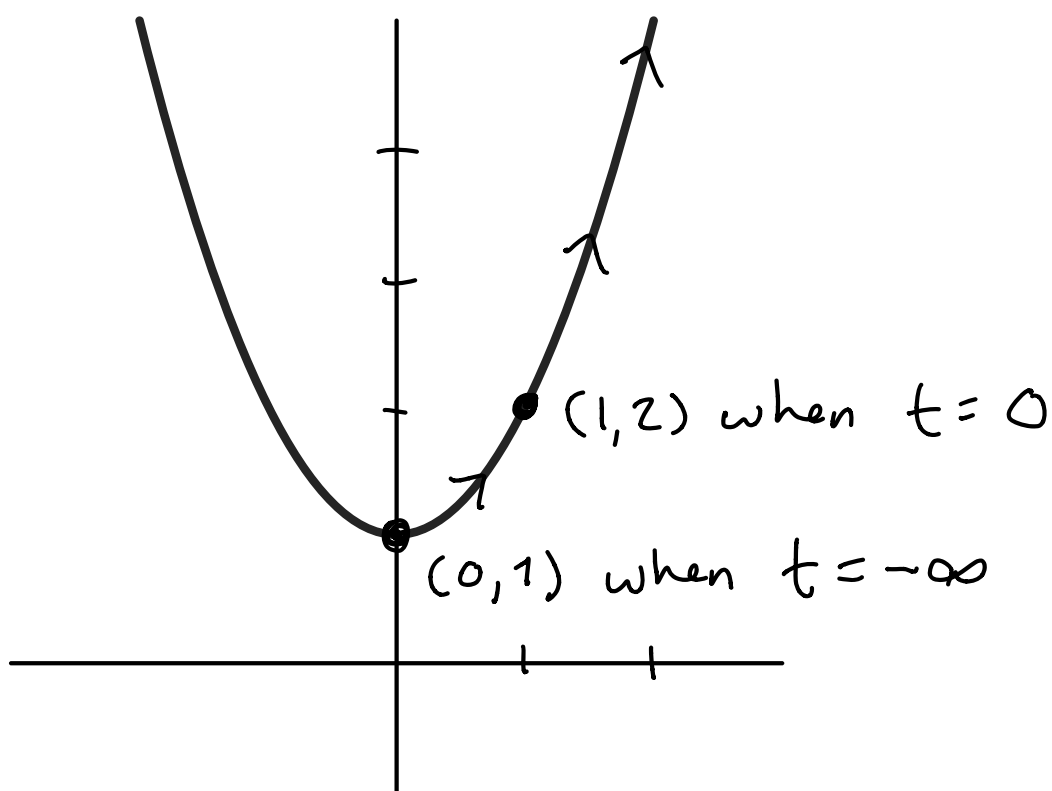
$$x = e^t$$

$$x^2 = (e^t)^2 = e^{2t}$$

$$x^2 + 1 = e^{2t} + 1 = y$$

$$y = x^2 + 1$$

Picture :



Actually we only get the right half of the parabola because $e^t > 0$ for all real values of t .

$$(c) \quad (x, y) = (1 + \cos t, 2 + \sin t)$$

$$x = 1 + \cos t$$

$$y = 2 + \sin t$$

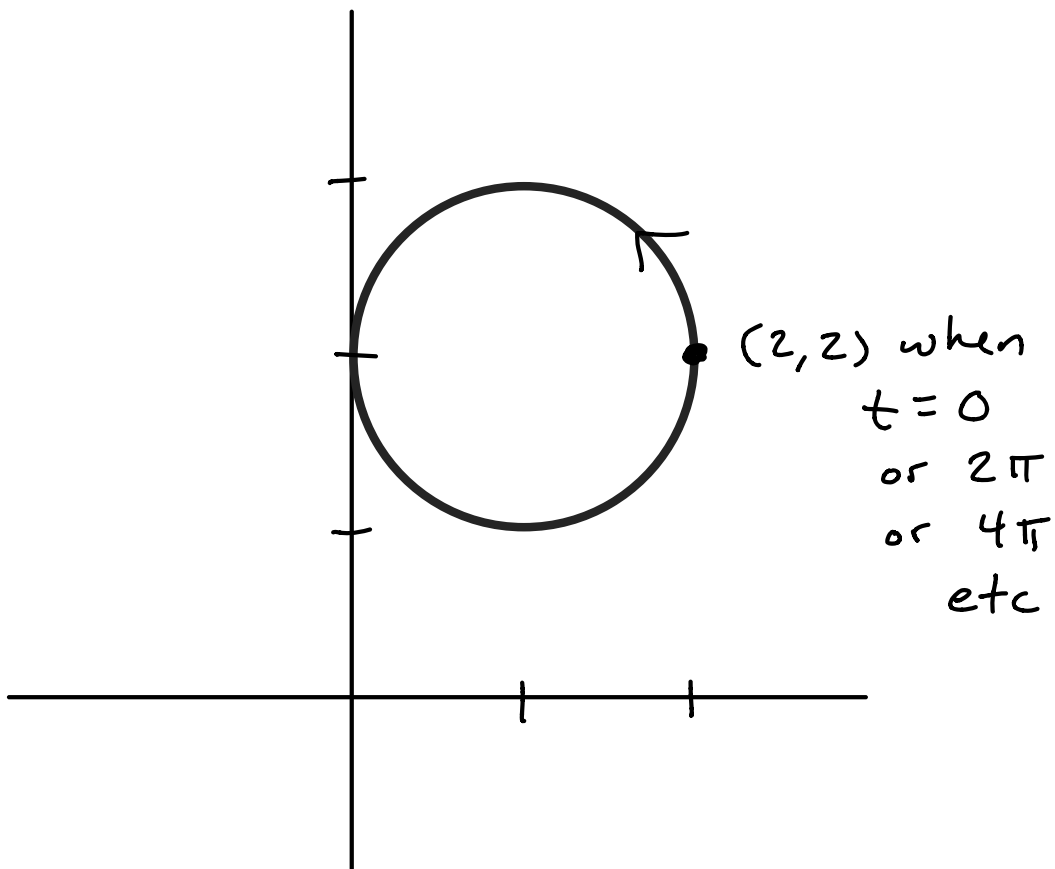
$$x - 1 = \cos t$$

$$y - 2 = \sin t$$

$$\text{So } (x-1)^2 + (y-2)^2 = \cos^2 t + \sin^2 t$$

$$(x-1)^2 + (y-2)^2 = 1.$$

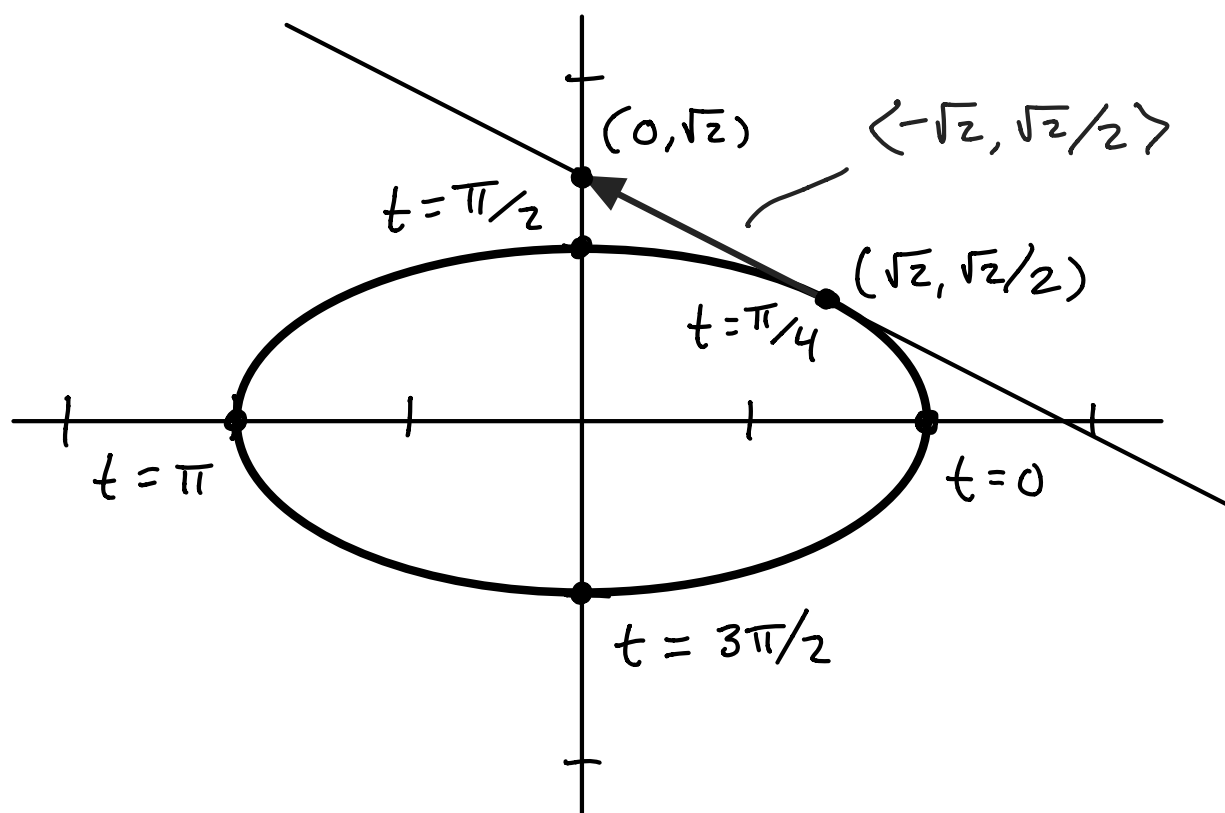
Picture :



2. A parametrized ellipse

$$\left(\frac{x}{2}\right)^2 + y^2 = 1$$

$$(x, y) = (2\cos t, \sin t)$$



velocity at time t :

$$\left\langle \frac{dx}{dt}, \frac{dy}{dt} \right\rangle = \langle -2\sin t, \cos t \rangle$$

speed at time t :

$$\sqrt{(-2\sin t)^2 + \cos^2 t} = \sqrt{4\sin^2 t + \cos^2 t}$$

At time $t = \pi/4$ we have

$$\text{position} = (\sqrt{2}, \sqrt{2}/2)$$

$$\text{velocity} = \langle -\sqrt{2}, \sqrt{2}/2 \rangle$$

The slope of the tangent line at time t is

$$\frac{dy}{dx} = \frac{dy/dt}{dx/dt} = \frac{\cos t}{-2 \sin t}$$

and at $t = \pi/4$ is

$$\frac{\cos(\pi/4)}{-2 \sin(\pi/4)} = \frac{\sqrt{2}/2}{-\sqrt{2}} = -\frac{1}{2}$$

Use point-slope to get the equation of the tangent line at $t = \pi/4$:

$$(y - y_0) = m(x - x_0)$$

$$y - \sqrt{2}/2 = -\frac{1}{2}(x - \sqrt{2})$$

$$y = -\frac{1}{2}x + \sqrt{2}$$

[There are several different ways to get this equation.]

The perimeter is the arc length between $t = 0$ and $t = 2\pi$:

$$\text{perimeter} = \int \text{speed } dt$$

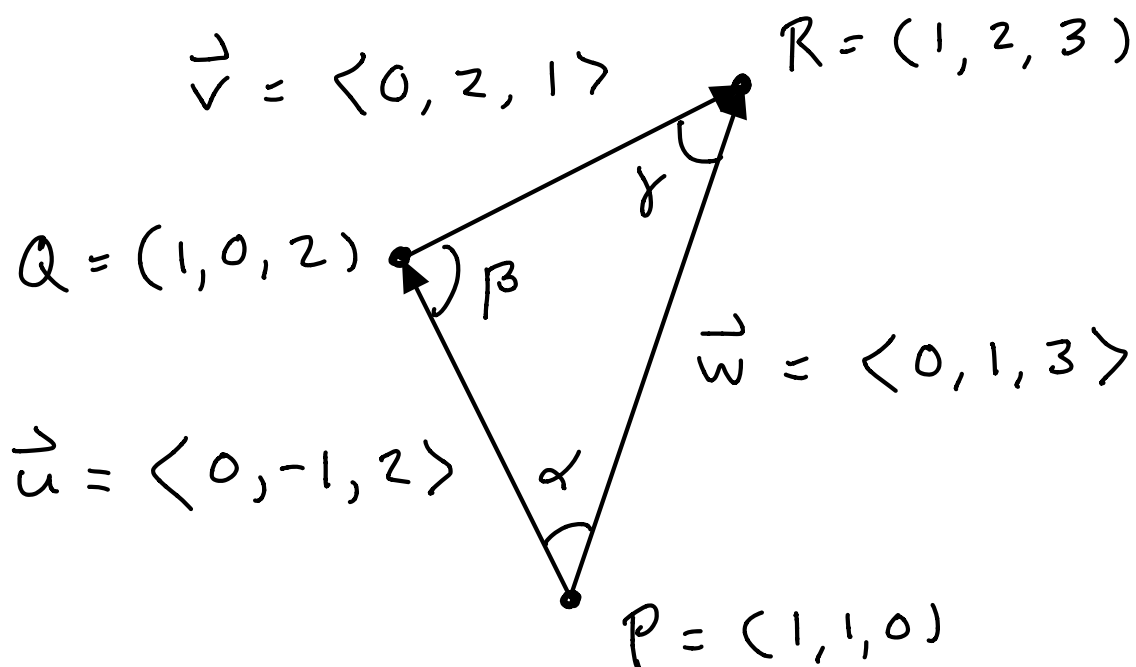
$$= \int_0^{2\pi} \sqrt{4\sin^2 t + \cos^2 t} \, dt$$

$$\approx 9.69$$

This integral cannot be solved by elementary means (the same is true for most arc lengths).

3. A triangle in space.

$$P = (1, 1, 0), \quad Q = (1, 0, 2), \quad R = (1, 2, 3)$$



[Why did I choose to draw it like this ?]

$$\|\vec{u}\| = \sqrt{\vec{u} \cdot \vec{u}} = \sqrt{0^2 + (-1)^2 + 2^2} = \sqrt{5}$$

$$\|\vec{v}\| = \sqrt{\vec{v} \cdot \vec{v}} = \sqrt{0^2 + 2^2 + 1^2} = \sqrt{5}$$

$$\|\vec{w}\| = \sqrt{\vec{w} \cdot \vec{w}} = \sqrt{0^2 + 1^2 + 3^2} = \sqrt{10}$$

$$\vec{u} \cdot \vec{w} = \|\vec{u}\| \|\vec{w}\| \cos \alpha$$

$$0 \cdot 0 + (-1) \cdot 1 + 2 \cdot 3 = \sqrt{5} \sqrt{10} \cos \alpha$$

$$\cos \alpha = 5 / \sqrt{5} \sqrt{10} = 1 / \sqrt{2}$$

$$\longrightarrow \alpha = \pi/4 = 45^\circ \text{ exactly!}$$

$$(-\vec{u}) \cdot \vec{v} = \|-\vec{u}\| \|\vec{v}\| \cos \beta$$

$$0 \cdot 0 + 1 \cdot 2 + (-2) \cdot 1 = \sqrt{5} \sqrt{5} \cos \beta$$

$$\cos \beta = 0$$

$$\longrightarrow \beta = 90^\circ$$

$$[\vec{u} \cdot \vec{v} = 0 \text{ so } \vec{u} \perp \vec{v} = 0]$$

$$(-\vec{v}) \cdot (-\vec{w}) = \|-\vec{v}\| \|-\vec{w}\| \cos \gamma$$

$$0 \cdot 0 + (-2)(-1) + (-1)(-3) = \sqrt{5} \sqrt{10} \cos \gamma$$

$$\cos \gamma = 5 / \sqrt{5} \sqrt{10} = 1/\sqrt{2}$$

$$\longrightarrow \gamma = \pi/4 = 45^\circ$$

[Actually we only needed to compute two of the angles because $\alpha + \beta + \gamma = \pi = 180^\circ$.]